

## AMALIY MASHG'ULOT-9

**Mavzu: C++ da oddiy jarayonlarni dasturlash. Chiziqli dasturlash texnologiyasi.**

### **Ishdan maqsad.**

Chiziqli algoritmlar va ular ustida amallar bajarish ko'nikmalarini oshirish. Dasturini tuzish.

**Nazariy qism.** Kiritish, chiqarish operatorlari, matematik funksiyalar.

Matematik funksiyalar C++ dasturlash tilida <math.h> kutubxonasida joylashgan bo'lib, ular quyidagicha yoziladi:

| №                                                     | Matematik ko'rinishi              | C++ dagi ko'rinishi       | Izoh                                                                                               |
|-------------------------------------------------------|-----------------------------------|---------------------------|----------------------------------------------------------------------------------------------------|
| <b>Trigonometrik funksiyalar</b>                      |                                   |                           |                                                                                                    |
| 1                                                     | $\sin x$                          | $\sin(x)$                 | $x$ radianga teng bo'lgan burchak sinusini qaytaradi                                               |
| 2                                                     | $\cos x$                          | $\cos(x)$                 | $x$ radianga teng bo'lgan burchak kosinusini qaytaradi                                             |
| 3                                                     | $\operatorname{tg} x$             | $\tan(x)$                 | $x$ radianga teng bo'lgan burchak tangensini qaytaradi                                             |
| 4                                                     | $\operatorname{ctg} x$            | $1/\tan(x)$               | $x$ radianga teng bo'lgan burchak kotangensini qaytaradi                                           |
| 5                                                     | $\sec x$                          | $\sinh(x)$                | $x$ radianga teng bo'lgan burchak giperbolik sinusini qaytaradi                                    |
| 6                                                     | $\operatorname{cosec} x$          | $\cosh(x)$                | $x$ radianga teng bo'lgan burchak giperbolik kosinusini qaytaradi                                  |
| <b>Teskari trigonometrik funksiyalar</b>              |                                   |                           |                                                                                                    |
| 1                                                     | $\arcsin x$                       | $\operatorname{asin}(x)$  | $[-1;1]$ chegarada berilgan $x$ ning arksinusini qaytaradi                                         |
| 2                                                     | $\arccos x$                       | $\operatorname{acos}(x)$  | $[-1;1]$ chegarada berilgan $x$ ning arkkosinusini qaytaradi                                       |
| 3                                                     | $\operatorname{arctg} x$          | $\operatorname{atan}(x)$  | $[-1;1]$ chegarada berilgan $x$ ning arktangensini qaytaradi                                       |
| 4                                                     | $\operatorname{arcctg} x$         |                           | $[-1;1]$ chegarada berilgan $x$ ning arkkotangensini qaytaradi                                     |
| <b>Daraja, ildiz, logarifm, modul, e natural soni</b> |                                   |                           |                                                                                                    |
| 1                                                     | $e^x$                             | $\exp(x)$                 | $e^x$ qiymatini qaytaradi                                                                          |
| 2                                                     | $a^x$                             | $\operatorname{pow}(a,x)$ | $a^x$ qiymatini qaytaradi                                                                          |
| 3                                                     | $ x $                             | $\operatorname{fabs}(x)$  | $x$ sonining absolyut qiymatini qaytaradi                                                          |
| 4                                                     | $\ln x$                           | $\log(x)$                 | $x$ sonining natural logarifmini qaytaradi                                                         |
| 5                                                     | $\log_{10} x$                     | $\log_{10}(x)$            | $x$ sonining 10 asosli logarifmini qaytaradi                                                       |
| 6                                                     | $\sqrt{x}$                        | $\operatorname{sqrt}(x)$  | $x$ sonining kvadrat ildizini qaytaradi                                                            |
| 7                                                     | $e = 2.718281828$                 | $M\_E$                    | $e$ sonining qiymatini qaytaradi.                                                                  |
| 8                                                     | $\pi = 3.141592653$               | $M\_PI$                   | $\pi$ sonining qiymatini qaytaradi                                                                 |
|                                                       |                                   |                           |                                                                                                    |
| 1                                                     | $x$ ga eng yaqin kichik butun son | $\operatorname{floor}(x)$ | $x$ haqiqiy qiymatni eng yaqin kichik butun songa aylantiradi va uni haqiqiy ko'rinishda qaytaradi |
| 2                                                     | $x$ ga eng yaqin katta butun son  | $\operatorname{ceil}(x)$  | $x$ haqiqiy qiymatni eng yaqin katta butun songa aylantiradi va uni haqiqiy ko'rinishda qaytaradi  |

|   |                                                   |                   |                                                                                              |
|---|---------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------|
| 3 | <i>a ni b ga bo'lgandagi qoldiq</i>               | <i>a%b</i>        | Butun a sonini butun b soniga bo'lgandagi qoldiqni butun ko'rinishda qaytaradi.              |
| 4 | <i>a/b bo'linmaning butun qismi</i>               | <i>a/b</i>        | Butun a sonini butun b soniga bo'lgandagi bo'linmaning butun qismini qaytaradi               |
| 5 | <i><math>c=\sqrt{a^2 + b^2}</math> gipotenuza</i> | <i>hypot(a,b)</i> | To'g'ri burchakli uchburchakning katetlari bo'yicha gipotenuzani hisoblab natijaga qaytaradi |

### Uslubiy ko'rsatmalar

#### Masalaning qo'yilishi.

Quyida berilgan chiziqli algoritmlarning dasturini tuzing.

$$A = \sin^2 \frac{(x-y^2)+tgx}{(x-y^2)+e^{3.7}} \sqrt{1 + lg^2|x - y^2|};$$

$$B = \frac{\cos(Ax) \sqrt[3]{A(x-y^2)}}{10^{-5}(Ax-4)+e^A}.$$

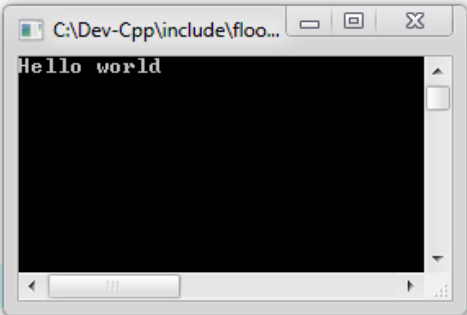
bu yerda  $x=1.27, y=0.4$ . Yechimini  $10^{-3}$  aniqlikda chiqaring. Natija:  $A=0.011$  va  $B=0.227$

C++ dasturlash muhitida ishlashga misollar.

1 – misol. 1 –dastur:

```

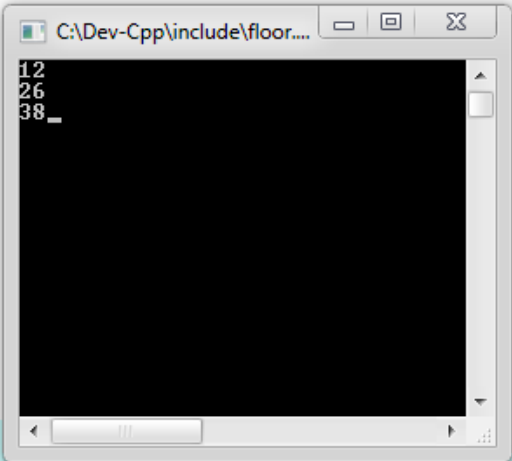
#include <iostream>
#include <math.h>
using namespace std;
int main() {
    cout<<"Hello world";
    system("pause>>void");
    return 0;
}
```



2 – misol. ikkita a va b sonini yig'indisini hisoblovchi dastur:

```

#include <iostream>
#include <math.h>
using namespace std;
int main() {
    int a,b;
    cin>>a>>b;
    cout<<a+b;
    system("pause>>void");
    return 0;
}
```



#### Berilgan masala uchun algoritm.

Yuqorida berilgan masalani hisoblash uchun birinchi navbatda uni matematik ko'rinishdan C++ dasturlash muhiti o'g'irish lozim.

$$A = \sin^2 \frac{(x-y^2)+tgx}{(x-y^2)+e^{3.7}} \sqrt{1 + lg^2|x - y^2|};$$

$$B = \frac{\cos(Ax) \sqrt[3]{A(x-y^2)}}{10^{-5}(Ax-4)+e^A}.$$

C++ muhitidagi ko'rinishi:

A=pow(sin(((x-y\*y)+tan(x))/((x-y\*y)+exp(3.7))),2)\*sqrt(1+pow(log10(fabs(x-y\*y)),2));  
B=cos(A\*x)\*pow(A\*(x-y\*y),1.0/3)/(0.00001\*(A\*x-4)+exp(A));

**Asosiy dastur matni:**

```
#include <iostream>
#include <math.h>
using namespace std;
int main(){
double A,B,x,y;
cin>>x>>y;
A=pow(sin(((x-y*y)+tan(x))/((x-y*y)+exp(3.7))),2)*sqrt(1+pow(log10(fabs(x-y*y)),2));
B=cos(A*x)*pow(A*(x-y*y),1.0/3)/(0.00001*(A*x-4)+exp(A));
cout.precision(3);
cout<<fixed<<A<<endl<<B;
system("pause>>void");
return 0;
}
```

#### **Berilgan misol ishlanish natijalari tahlili.**

1. **#include** – kutubxonalarni qo'shish direktivasi;
2. **<iostream>** - kiritish chiqarish kutubxonasi;
3. **<math.h>** - matematik funksiyalar berilgan kutubxona;
4. **using namespace std** – std nomlar fazosidan standart oqimda kiritish, chiqarishda foydalaniladi;

**int main() {**

...

**return 0;}** - asosiy funksiya dastur sintaksis to'g'ri yozilgan bo'lsa funksiya return orqali 0 qiymatni qaytaradi.

5. **double A, B, x, y** - haqiqiy turga mansub 4 ta o'zgaruvchi uchun xotiradan joy ajratadi.
6. **cin>>x>>y** – xotiradan ajratilgan x va y nomli haqiqiy o'zgaruvchilarga klaviaturadan qiymat kiritiladi.
7. **A=pow(sin(((x-y\*y)+tan(x))/((x-y\*y)+exp(3.7))),2)\*sqrt(1+pow(log10(fabs(x-y\*y)),2));**  
xotiradan ajratilgan A nomli o'zgaruvchiga yuqoridagi funksiya qiymati ta'minlanadi.
8. **B=cos(A\*x)\*pow(A\*(x-y\*y),1.0/3)/(0.00001\*(A\*x-4)+exp(A));** - xotiradan ajratilgan B nomli o'zgaruvchiga yuqoridagi funksiya qiymati ta'minlanadi.
9. **cout.precision(3); cout<<fixed<<A<<endl<<B** – A va B o'zgaruvchilarda mavjud qiymatlarni ekranga  $10^{-3}$  aniqlikda yaxlitlab chiqaradi.
10. **system("pause>>void")** - natijani ekranda ushlab turish uchun foydalaniladi.

**Natijalar listingi.**



**1-masala.** A va B ikkita haqiqiy sonlar berilgan. Ularning yig'indisi, ayirmasi va ko'paytmasini hisoblang.

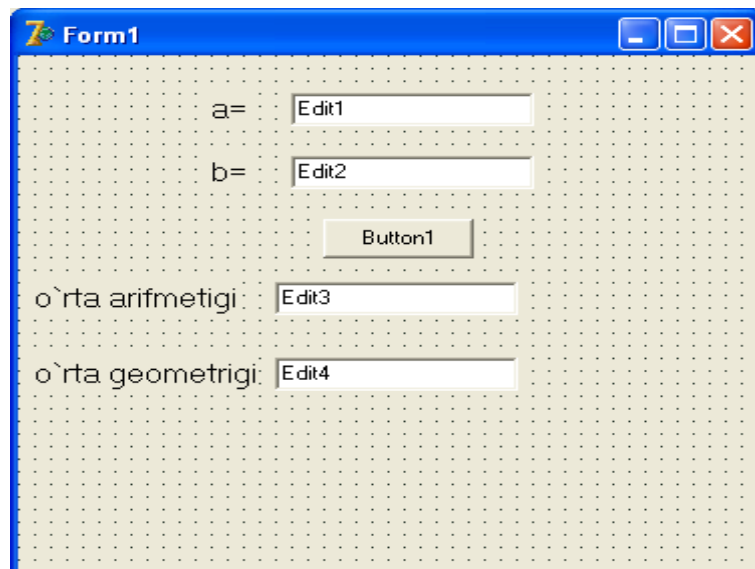
Yechish. A va b sonlar yig'indisini S, ayirmasini D, ko'paytmasini K bilan belgilasak,  $S=a+b$ ,  $d=a-b$ ,  $k=a*b$  formulalar o'rinli bo'ladi.

*Borland C++ Builder da dasturu:*

```
void __fastcall TForm1::Button1Click(TObject *Sender)
{
    float a,b,s,p;
    a=StrToFloat(Edit1->Text);
    b=StrToFloat(Edit2->Text);
    s=a+b;
    p=a*b;
    Edit3->Text=FloatToStr(s);
    Edit4->Text=FloatToStr(p);
}
```

**2-masala.** Ikkita musbat son berilgan, bu sonlarning o'rta arifmetik va o'rta geometrik qiymatlarini aniqlang.

Yechish. A va b sonlarning o'rta arifmetik qiymatini c, o'rta geometrik qiymatini B bilan belgilasak,  $c = \frac{a+b}{2}$ ;  $d = \sqrt{a \cdot b}$ ; formulalar o'rinli.



*Borland C++ Builder da dasturu:*

```
#include <math.h>
```

```
void __fastcall TForm1::Button1Click(TObject *Sender)
```

```
{
```

```
float a,b,s,p;
```

```
a=StrToFloat(Edit1->Text);
```

```
b=StrToFloat(Edit2->Text);
```

```
s=(a+b)/2;
```

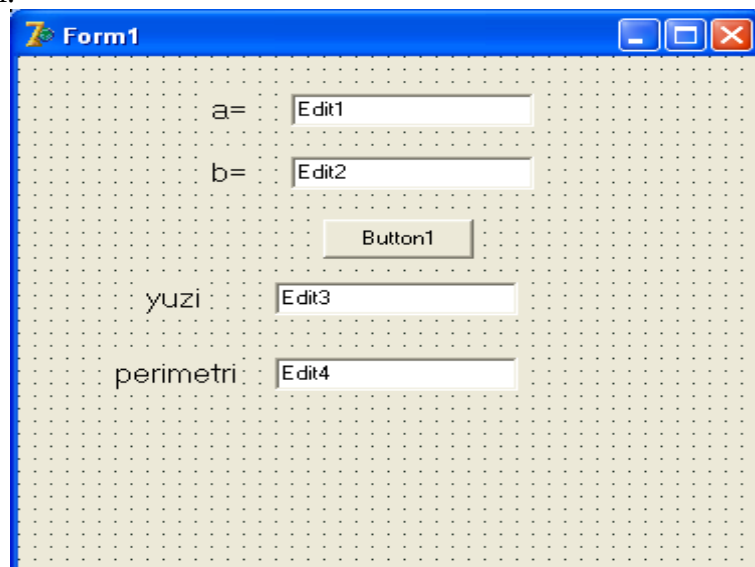
```
p=sqrt(a*b);
```

```
Edit3->Text=FloatToStr(s);
```

```
Edit4->Text=FloatToStr(p);
```

```
}
```

**3-masala.** Tomonlari A va B ga teng to'g'ri to'rtburchakning yuzi va perimetri hisoblansin. Yechish. To'g'ri to'rtburchakning yuzi  $s = a \cdot b$ , perimetri  $p = 2 \cdot (a + b)$  formulalar yordamida aniqlanadi.



*Borland C++ Builder da dasturu:*

```
void __fastcall TForm1::Button1Click(TObject *Sender)
```

```
{
```

```
float a,b,s,p;
```

```
a=StrToFloat(Edit1->Text);
```

```

b=StrToFloat(Edit2->Text);
s=a+b;
p=2*(a*b);
Edit3->Text=FloatToStr(s);
Edit4->Text=FloatToStr(p);
}

```

**4-masala.** R1, R2, R3 uchta qarshiliklar ketma-ket ulangan zanjirning qarshiligini aniqlang. Yechish.

Zanjirning umumiy qarshiligi R bilan belgilasak, ketma-ket ulashda  $R = R_1 + R_2 + R_3$  formulalar o'rinli bo'ladi.

*Borland C++ Builder da dasturu:*

```

void __fastcall TForm1::Button1Click(TObject *Sender)
{
float R1,R2,R3,R;
R1=StrToFloat(Edit1->Text);
R2=StrToFloat(Edit2->Text);
R3=StrToFloat(Edit3->Text);
R=R1+R2+R3;
Edit4->Text=FloatToStr(R);
}

```

**5-masala.** Massalari M1 va M2 (kg) ga teng, oralaridagi masofa R (m) ga teng bo'lgan ikkita jismlarning o'zaro tortilish kuchi F aniqlansin. Bunda gravitatsion doimiy  $G=6,672 \cdot 10^{-11}$  (N·m<sup>2</sup>/kg<sup>2</sup>) deb olinsin.

Yechish. Butun olam tortilish qonuniga ko'ra  $F = G \frac{m_1 \cdot m_2}{R^2}$ ;

Erning massasi  $m_1 = 5,97 \cdot 10^{24}$

Oyning massasi  $m_2 = 7,35 \cdot 10^{22}$

Er bilan Oy orasidagi masofa  $R = 3,844 \cdot 10^8$

Izoh. Er bilan Oyning massalari kilogrammda, masofa merta, kuch Nyutonda o'lchanadi.

*Borland C++ Builder da dasturu:*

```
void __fastcall TForm1::Button1Click(TObject *Sender)
{
    float M1,M2,R,F;
    M1=StrToFloat(Edit1->Text);
    R2=StrToFloat(Edit2->Text);
    R=StrToFloat(Edit3->Text);
    F=G*M1*M2/(R*R);
    Edit4->Text=FloatToStr(M);
}
```

**6-masala.** Teng tomonli uchburchakning tomoni A ga teng. Uchburchakning yuzini toping.

Yechish. Teng tomonli uchburchakning yuzini S bilan belgilasak,  $S = a^2 \cdot \frac{\sqrt{3}}{4}$

Formula o'rinli bo'ladi.

*Borland C++ Builder da dasturu:*

```
#include <math.h>
void __fastcall TForm1::Button1Click(TObject *Sender)
{
    float a,s;
    a=StrToFloat(Edit1->Text);
    S=(sqrt(3))*a*a/4;
    Edit2->Text=FloatToStr(s);
}
```

**7-masala.** Koordinatalari  $X_1, Y_1$  va  $X_2, Y_2$  ga teng bo'lgan nuqtalari orasidagi masofani hisoblang.

Yechish. Ikki nuqta orasidagi masofa  $S = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ ; formula yordamida aniqlanadi.

*Borland C++ Builder da dasturi:*

```
#include <math.h>
void __fastcall TForm1::Button1Click(TObject *Sender)
{
    float x1,x2,y1,y2,d;
    x1=StrToFloat(Edit1->Text);
    x2=StrToFloat(Edit2->Text);
    y1=StrToFloat(Edit3->Text);
    y2=StrToFloat(Edit4->Text);
    d=sqrt(sqr(x2-x1)+sqr(y2-y1));
    Edit5->Text=FloatToStr(d);
}
```

**8-masala.** Birinchi hadi  $A$ , ayirmasi  $D$ , hadlari soni  $N$  ga teng arifmetik progressiyaning hadlarining yig'indisini hisoblang.

Yechish. Arifmetik progressiya istalgan hadi va hadlari yig'indisi uchun  $a_n = a + d \cdot (n - 1)$ ,  $S_n = \frac{2 \cdot a + d \cdot (n - 1) \cdot n}{2}$ , formulalar o'rinli bo'ladi.



*Borland C++ Builder da dasturu:*

```
void __fastcall TForm1::Button1Click(TObject *Sender)
{
    float a1,d,n,S;
    a1=StrToFloat(Edit1->Text);
    d=StrToFloat(Edit2->Text);
    n=StrToFloat(Edit3->Text);
    S=(2*a1+d*(n-1))/2*n;
    Edit4->Text=FloatToStr(S);
}
```

**9-masala.** Birinchi hadi B, maxraji Q va hadlari soni N ga teng geometrik progressiyaning hadlarining yig'indisini hisoblang.

Yechish.

Geometrik progressiyaning istalgan hadi va hadlari yig'indisi

$b_n = b \cdot q^{n-1}$ ;  $s_n = \frac{b \cdot q - b}{q - 1}$ ; formula yordamida aniqlanadi.

*Borland C++ Builder da dasturu:*

```
void __fastcall TForm1::Button1Click(TObject *Sender)
{
    float B,q,n,S;
```

```

B=StrToFloat(Edit1->Text);
q=StrToFloat(Edit2->Text);
n=StrToFloat(Edit3->Text);
S=(B*(1-exp(N*ln(q))))/(1-q);
Edit4->Text=FloatToStr(S);
}

```

**10-masala.** Uchta idishga suv solingan. Idishlardagi suvning temperaturasi  $T_1$ ,  $T_2$ ,  $T_3$  ga teng, hajmi  $V_1$ ,  $V_2$ ,  $V_3$  (l) ga teng. Idishlardagi suvni bitta idishga quyilsa, uning hajmi va temperaturasi qanday bo'ladi?

Yechish. Idishlardagi suvni bitta idishga quyilsa, suvning hajmi va temperaturasi

$V = V_1 + V_2 + V_3$ ;  $T = \frac{V_1 * T_1 + V_2 * T_2 + V_3 * T_3}{V}$  formulalar bilan aniqlanadi.

*Borland C++ Builder da dasturu:*

```

void __fastcall TForm1::Button1Click(TObject *Sender)
{
float T1,T2,T3,T,V1,V2,V3,V;
T1=StrToFloat(Edit1->Text);
T2=StrToFloat(Edit2->Text);
T3=StrToFloat(Edit3->Text);
V1=StrToFloat(Edit4->Text);
V2=StrToFloat(Edit5->Text);
V3=StrToFloat(Edit6->Text);
T=(T1*V1+T2*V2+T3*V3)/(V1+V2+V3);
V=V1+V2+V3;
Edit7->Text=FloatToStr(T);
Edit8->Text=FloatToStr(V);
}

```